



DEALER BEST PRACTICE SERIES

Tool to Repair Damaged OHT Exhaust Pipe

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

**MARC
Management**

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June 2011
1011-1.04-1218

1.0 Introduction

Special tools that enable fast and simple repairs are an important element of a mine site's Repair Management discipline. Tools for specific tasks promote efficient repair execution and help minimize machine downtime, improve machine repair time, and put the truck back to work faster to help production.

Denting, crushing, or other damage to the exhaust pipes on Cat Off Highway Trucks can happen during truck loading by rocks falling out the bucket or off the canopy. When this occurs, a means of easily and quickly repairing the exhaust by straightening it can minimize truck downtime.

2.0 Best Practice Description

This special tool allows quick and simple repairs to damaged or dented exhaust pipes.

3.0 Implementation Steps

1. Identify machine and size of exhaust that needs straightening.
2. Measure the inside diameter of an exhaust pipe that is not damaged.
3. Research local scissor jack suppliers for a suitable jack size. Note: Be sure to ensure the closed height of the jack is smaller than the inside diameter of the exhaust pipe.
 - a. For example: 793D exhaust extension 8X-3142, has an inside diameter of 260 mm. The collapsed height of the jack + two times the thickness of the steel plates needs to be less than 260 mm in order to insert the jack. Additional clearance is advisable to allow space to insert the jack into the damaged exhaust pipe.
4. Purchase appropriately sized scissor-jack (Capacity of 0.75–1.0 ton jack is sufficient).
 - a. For example:
 - i. Manufacturer: US General #66907
 - ii. Manufacturer: Torin #T10152
 - iii. Manufacturer: Torin #T144871
5. Weld correct sized steel plates that conform to the ID of the exhaust pipe. See attached drawing.
6. Insert scissor-jack into damaged exhaust pipe.
7. Extend scissor-jack to bend damaged exhaust to correct shape.

Damaged exhaust on LOHT:



Scissor-jack tool with plates welded to jack.

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Install tool in exhaust and extend to straighten. Photo of repaired exhaust pipe.



4.0 Benefits

Scissor-jack design with conforming steel plates to the exhaust ID offers a simple, inexpensive, and quick method of straightening a damaged exhaust pipe.

5.0 Resources Required

The major resource required to implement this Best Practice is purchasing a scissor jack welding the steel plates to the jack.

Fabrication costs are approximately **US \$ 300** / per set.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

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NUMBER
1011-1.04-1218

8.0 Acknowledgements

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